

Marginal expectation and free-energy additivity for separable Gibbs measures

Tide retrospective, laplace seabed

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1 Setting

The laplace seabed’s separable-potential module `TwoD/AddSeparable.lean` establishes the factorisation facts for a product potential $L(x, y) = U(x) + V(y)$: the partition function factors, $Z_{2D} = Z_U \cdot Z_V$; separable observables factor in expectation; and the mixed covariance vanishes. A recent companion, `SeparableSameCoordCov`, added the same-coordinate covariance inheritance at the second-cumulant level. This tide fills in the two remaining elementary facts of the separable picture — one at the mean level, one at the free-energy (CGF) level — both of which follow directly from the factorisation already in hand. It is a deliberate consolidation step rather than a new frontier: the monomial cumulant ladder and the block-diagonal covariance structure being complete, this rounds out the separable module’s basic API.

2 The thing formalised

For a separable potential and both marginal partition functions nonzero, a single-coordinate observable marginalises to its one-dimensional Gibbs expectation,

$$\langle f \circ \text{fst} \rangle_{2D, t} = \langle f \rangle_{U, t}, \quad \langle g \circ \text{snd} \rangle_{2D, t} = \langle g \rangle_{V, t},$$

the mean-level companion of the covariance-level `SeparableSameCoordCov`. And the free energy is additive,

$$\log Z_{2D}(t) = \log Z_U(t) + \log Z_V(t),$$

the log-partition (cumulant-generating-function) reading of the partition-function factorisation. The Lean names are `gibbsExpectation_addSeparable_fst` / `_snd` and `logPartition_addSeparable`, in a new leaf file `TwoD/SeparableMarginal.lean`.

3 Proof strategy

The marginal expectation is the separable-observable factorisation `gibbsExpectation_separable_addSeparable` run with the constant 1 in the free coordinate: $\langle f(z_1) \rangle_{2D} = \langle f \rangle_U \cdot \langle 1 \rangle_V = \langle f \rangle_U$, since the Gibbs expectation of the constant 1 is 1 (`gibbsExpectation_const`). This is exactly the collapse that was previously *inlined* inside the cross-covariance-vanishing proof; the tide lifts it to a named, reusable lemma. The free-energy additivity is one rewrite: the partition-function factorisation followed by $\log(ab) = \log a + \log b$ (`Real.log_mul`, whose two nonzero-base side conditions are discharged by the positivity of each marginal partition function).

4 Roadblocks and resolutions

None — both results compiled on the first attempt, and the full library rebuilt clean. As with the sibling covariance tide, the only process choice was to place the additions in a new leaf file importing `AddSeparable` rather than editing the core hub, so that no downstream file re-elaborates and no latent missing-header warnings cascade. The build is warning-clean and axiom-clean (the three standard axioms only).

5 What was Mathlib, what was new

The only Mathlib input is `Real.log_mul`. Everything else is the seabed’s own separable-factorisation API. What is new is the exposure of the marginal expectation as a first-class lemma (previously reachable only by re-deriving the inlined step) and the CGF-level statement of free-energy additivity, completing the elementary separable-potential toolkit at the mean and free-energy levels alongside the existing partition, expectation, and covariance facts. No GPT consult was needed.